# MATH 2111

## Tutorial 3: Coordinates, Dimension and Rank

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# Coordinates and Ordered Bases

## Unique representation

Let $\mathcal{B} = (\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n)$ be an ordered basis of a vector space $V$. Every $\boldsymbol{x} \in V$ has a unique representation

$$\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n,$$

and its coordinate vector is

$$[\boldsymbol{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 & \cdots & c_n \end{bmatrix}^T.$$

The coordinate mapping $\boldsymbol{x} \mapsto [\boldsymbol{x}]_{\mathcal{B}}$ is a one-to-one linear transformation from $V$ onto $\mathbb{R}^n$.

## Coordinates in $\mathbb{R}^n$

For

$$P_{\mathcal{B}} = [\boldsymbol{b}_1 \ \cdots \ \boldsymbol{b}_n],$$

we have

$$\boldsymbol{x} = P_{\mathcal{B}}[\boldsymbol{x}]_{\mathcal{B}}, \qquad [\boldsymbol{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\boldsymbol{x}.$$

Thus $P_{\mathcal{B}}$ maps $\mathcal{B}$-coordinates to standard coordinates.

# Dimension and Basis Selection

## Dimension

If $V$ has a finite basis, then

$$\dim V = \text{number of vectors in any basis of } V.$$

Also,

$$\dim\{\mathbf{0}\} = 0.$$

A space with no finite spanning set is infinite-dimensional. If $\dim V = p$, every set of more than $p$ vectors in $V$ is linearly dependent.

## Two useful selection principles

**Basis Theorem:** In a $p$-dimensional space, exactly $p$ linearly independent vectors form a basis; exactly $p$ spanning vectors also form a basis.

**Spanning Set Theorem:** A vector that is a linear combination of the remaining generators can be removed without changing the span.

If $H \leq V$ and $V$ is finite-dimensional, then

$$\dim H \leq \dim V,$$

and every linearly independent set in $H$ can be extended to a basis of $H$.

# Row Space, Column Space and Rank

Let $A$ be an $m \times n$ matrix.

## How to obtain bases

- $\operatorname{Row} A \subseteq \mathbb{R}^n$: use the nonzero rows of an echelon form of $A$;
- $\operatorname{Col} A \subseteq \mathbb{R}^m$: use the pivot columns of the original matrix $A$;
- $\operatorname{Nul} A \subseteq \mathbb{R}^n$: write the general solution of $A\boldsymbol{x} = 0$ using the free variables.

Row-equivalent matrices have the same row space, but row operations generally change the column space.

## Rank Theorem

$$\operatorname{rank} A = \dim \operatorname{Col} A = \dim \operatorname{Row} A = \#\{\text{pivots}\},$$

$$\boxed{\operatorname{rank} A + \dim \operatorname{Nul} A = n}.$$

Moreover,

$$\operatorname{rank} A^T = \operatorname{rank} A.$$

For an $n \times n$ matrix,

$$\operatorname{rank} A = n$$

is equivalent to invertibility, trivial null space, linearly independent columns, spanning columns, and the associated map being both one-to-one and onto.

# Problem 1: Coordinate Conversion

## Problem 1

In $\mathbb{R}^2$, let

$$\mathcal{B} = \left( \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right), \qquad \mathcal{C} = \left( \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right).$$

Suppose

$$[\boldsymbol{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

(a) Find $\boldsymbol{v}$ in standard coordinates.

(b) Derive the matrix $P_{\mathcal{C} \leftarrow \mathcal{B}}$ satisfying

$$[\boldsymbol{x}]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}} [\boldsymbol{x}]_{\mathcal{B}}.$$

(c) Find $[\boldsymbol{v}]_{\mathcal{C}}$ and interpret the columns of $P_{\mathcal{C} \leftarrow \mathcal{B}}$.

# Solution to Problem 1: Standard Coordinates

The basis matrices are

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad P_{\mathcal{C}} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Since $P_{\mathcal{B}}$ maps $\mathcal{B}$-coordinates to standard coordinates,

$$\boldsymbol{v} = P_{\mathcal{B}}[\boldsymbol{v}]_{\mathcal{B}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \boxed{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}.$$

Equivalently,

$$\boldsymbol{v} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

# Solution to Problem 1: Changing the Basis

For every $\boldsymbol{x} \in \mathbb{R}^2$,

$$P_{\mathcal{C}}[\boldsymbol{x}]_{\mathcal{C}} = \boldsymbol{x} = P_{\mathcal{B}}[\boldsymbol{x}]_{\mathcal{B}}.$$

Hence

$$[\boldsymbol{x}]_{\mathcal{C}} = P_{\mathcal{C}}^{-1} P_{\mathcal{B}}[\boldsymbol{x}]_{\mathcal{B}},$$

so

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = P_{\mathcal{C}}^{-1} P_{\mathcal{B}} = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix}.$$

Therefore

$$[\boldsymbol{v}]_{\mathcal{C}} = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \boxed{\begin{bmatrix} -2 \\ 3 \end{bmatrix}}.$$

The columns of $P_{\mathcal{C}\leftarrow\mathcal{B}}$ are

$$[\boldsymbol{b}_1]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad [\boldsymbol{b}_2]_{\mathcal{C}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

# Problem 2: Coordinates in a Polynomial Space

## Problem 2

In $P_2$, use the ordered basis

$$\mathcal{B} = (1+t,\ t+t^2,\ 1+t^2).$$

Let

$$p_1(t) = 2 + t + t^2,$$
$$p_2(t) = 1 + t + 2t^2,$$
$$p_3(t) = 3 + 2t + 3t^2.$$

(a) Find $[p_1]_{\mathcal{B}}$, $[p_2]_{\mathcal{B}}$ and $[p_3]_{\mathcal{B}}$.

(b) Find a linear relation among $p_1, p_2, p_3$, and give a basis for

$$H = \mathrm{Span}\{p_1, p_2, p_3\}.$$

(c) Determine whether $1 + t$ belongs to $H$.

# Solution to Problem 2: Coordinate Vectors

Writing each polynomial as a linear combination of the ordered basis gives

$$p_1 = (1 + t) + (1 + t^2),$$

$$p_2 = (t + t^2) + (1 + t^2),$$

$$p_3 = (1 + t) + (t + t^2) + 2(1 + t^2).$$

Thus

$$[p_1]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad [p_2]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad [p_3]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Since the coordinate mapping is an isomorphism, linear relations among the polynomials are exactly the same as linear relations among these coordinate vectors.

# Solution to Problem 2: Dependence and Membership

The coordinate vectors satisfy

$$[p_3]_{\mathcal{B}} = [p_1]_{\mathcal{B}} + [p_2]_{\mathcal{B}}.$$

Hence

$$\boxed{p_3 = p_1 + p_2}.$$

The first two coordinate vectors are linearly independent, so

$$\boxed{(p_1, p_2) \text{ is a basis for } H}, \qquad \dim H = 2.$$

Now

$$[1+t]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

If $1 + t = ap_1 + bp_2$, then

$$a \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

The first two entries force $a = 1$ and $b = 0$, but then the third entry is $1$, not $0$. Therefore

$$\boxed{1 + t \notin H}.$$

# Problem 3: Dimension and Basis Theorems

## Problem 3

Let

$$H = \{p \in P_3 : p(1) = 0\},$$

and define

$$q_1 = t - 1, \qquad q_2 = t^2 - 1, \qquad q_3 = t^3 - 1.$$

(a) Show that $(q_1, q_2, q_3)$ is a basis for $H$, and find $\dim H$.

(b) From

$$S = \{q_1, q_2, q_3, q_1 + q_2\},$$

remove a redundant vector without changing the span.

(c) Use the Basis Theorem to decide whether

$$\mathcal{R} = (q_1 + q_2,\ q_2 + q_3,\ q_1 + q_3)$$

is a basis for $H$.

# Solution to Problem 3: A Basis for $H$

Let

$$p(t) = a + bt + ct^2 + dt^3 \in P_3.$$

The condition $p(1) = 0$ gives

$$a + b + c + d = 0, \qquad a = -b - c - d.$$

Therefore

$$p(t) = b(t-1) + c(t^2 - 1) + d(t^3 - 1) = bq_1 + cq_2 + dq_3.$$

Thus $q_1, q_2, q_3$ span $H$. They are linearly independent because a relation

$$\alpha q_1 + \beta q_2 + \gamma q_3 = 0$$

forces the coefficients of $t, t^2, t^3$ to be zero. Hence

$$\boxed{(q_1, q_2, q_3) \text{ is a basis for } H}, \qquad \boxed{\dim H = 3}.$$

# Solution to Problem 3: Selecting a Basis

Since $q_1 + q_2$ is a linear combination of the other vectors in $S$, the Spanning Set Theorem gives

$$\operatorname{Span} S = \operatorname{Span}\{q_1, q_2, q_3\} = H.$$

For $\mathcal{R}$, use coordinates relative to $(q_1, q_2, q_3)$:

$$M = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}.$$

There are three pivots, so the three vectors in $\mathcal{R}$ are linearly independent. Since $\dim H = 3$, the Basis Theorem gives

$$\boxed{\mathcal{R} \text{ is a basis for } H}.$$

# Problem 4: Intersections of Subspaces

## Problem 4

In $\mathbb{R}^3$, let

$$H = \text{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\},$$

$$K = \left\{ \boldsymbol{x} \in \mathbb{R}^3 : x_1 + 2x_2 - x_3 = 0 \right\}.$$

(a) Write $H$ as the solution set of one homogeneous equation. State $\dim H$ and $\dim K$ and interpret the two subspaces geometrically.

(b) Find a basis for $H \cap K$.

(c) Extend that basis to a basis for $H$ and to a basis for $K$.

# Solution to Problem 4: Equations and Dimensions

A vector in $H$ has the form

$$a\begin{bmatrix}1\\1\\0\end{bmatrix}+b\begin{bmatrix}0\\1\\1\end{bmatrix}=\begin{bmatrix}a\\a+b\\b\end{bmatrix}.$$

Therefore

$$\boxed{H=\{\boldsymbol{x}:x_1-x_2+x_3=0\}}.$$

The two given generators of $H$ are linearly independent, so $\dim H=2$.
For $K$,

$$x_1=-2x_2+x_3,$$

so

$$K=\operatorname{Span}\left\{\begin{bmatrix}-2\\1\\0\end{bmatrix},\begin{bmatrix}1\\0\\1\end{bmatrix}\right\},\qquad \dim K=2.$$

Thus both $H$ and $K$ are planes through the origin in $\mathbb{R}^3$.

# Solution to Problem 4: The Intersection

A vector in $H \cap K$ satisfies

$$\begin{cases} x_1 - x_2 + x_3 = 0, \\ x_1 + 2x_2 - x_3 = 0. \end{cases}$$

Adding and subtracting the two equations gives

$$3x_2 - 2x_3 = 0, \qquad x_1 = x_2 - x_3.$$

Taking $x_2 = 2s$ gives $x_3 = 3s$ and $x_1 = -s$. Hence

$$\boldsymbol{x} = s \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix},$$

so

$$\boxed{H \cap K = \operatorname{Span}\left\{ \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} \right\}}, \qquad \boxed{\dim(H \cap K) = 1}.$$

Thus the intersection is a line through the origin.

# Solution to Problem 4: Extending the Basis

Let

$$\boldsymbol{g} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}.$$

Since $\boldsymbol{g}$ is not a multiple of either added vector, one extension to a basis for $H$ is

$$\boxed{\left( \boldsymbol{g}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right)}.$$

Similarly, one extension to a basis for $K$ is

$$\boxed{\left( \boldsymbol{g}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right)}.$$

Both extensions contain two linearly independent vectors in a two-dimensional space. In particular,

$$\dim(H \cap K) = 1 \leq 2 = \dim H = \dim K.$$

# Problem 5: Row, Column and Null Spaces

## Problem 5

Let

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -1 \\ 1 & 2 & 1 & 1 & 2 \end{bmatrix}.$$

(a) Find $\mathrm{rref}(A)$ and identify the pivot columns.

(b) Find bases for $\mathrm{Row}\, A$, $\mathrm{Col}\, A$ and $\mathrm{Nul}\, A$.

(c) Find $\mathrm{rank}\, A$ and $\dim \mathrm{Nul}\, A$, and verify the Rank Theorem.

(d) Explain why the row-space basis may be taken from the reduced matrix, while the column-space basis must be taken from the original matrix.

# Solution to Problem 5: Reduction and Pivots

The third row is the sum of the first two rows. The successive replacements

$$R_3 \leftarrow R_3 - R_1, \qquad R_3 \leftarrow R_3 - R_2$$

give

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are columns $1$ and $3$.
The nonzero rows of the reduced matrix form a basis for the row space:

$$\boxed{\operatorname{Row} A = \operatorname{Span}\left\{ \begin{bmatrix} 1 & 2 & 0 & -1 & 3 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 2 & -1 \end{bmatrix} \right\}.}$$

Here $\operatorname{Row} A \subseteq \mathbb{R}^5$.

# Solution to Problem 5: Column and Null Spaces

The pivot columns must be taken from the original matrix:

$$\boxed{\operatorname{Col} A = \operatorname{Span}\left\{\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}\right\}.}$$

Here $\operatorname{Col} A \subseteq \mathbb{R}^3$.
For $A\boldsymbol{x} = 0$,

$$\begin{cases} x_1 + 2x_2 - x_4 + 3x_5 = 0, \\ x_3 + 2x_4 - x_5 = 0. \end{cases}$$

Taking $x_2, x_4, x_5$ free gives

$$\boldsymbol{x} = x_2 \begin{bmatrix}-2\\1\\0\\0\\0\end{bmatrix} + x_4 \begin{bmatrix}1\\0\\-2\\1\\0\end{bmatrix} + x_5 \begin{bmatrix}-3\\0\\1\\0\\1\end{bmatrix}.$$

Thus these three vectors form a basis for $\operatorname{Nul} A \subseteq \mathbb{R}^5$.

# Solution to Problem 5: Rank–Nullity

There are two pivot positions, so

$$\boxed{\operatorname{rank} A = 2}.$$

There are three free variables, so

$$\boxed{\dim \operatorname{Nul} A = 3}.$$

Since $A$ has five columns,

$$\operatorname{rank} A + \dim \operatorname{Nul} A = 2 + 3 = 5,$$

which verifies the Rank Theorem.

**Why the extraction rules differ**

Row operations replace rows by linear combinations of rows, and the operations are reversible. Therefore they preserve the entire row space.
Row operations do not preserve the actual columns or the column space. They preserve only which column positions are pivot positions. Hence a basis for $\operatorname{Col} A$ must use the corresponding columns of the original matrix.

# Problem 6: Rank and Invertibility

## Problem 6

For $\lambda \in \mathbb{R}$, let

$$A_\lambda = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & \lambda \end{bmatrix}.$$

(a) Determine $\operatorname{rank} A_\lambda$ and $\dim \operatorname{Nul} A_\lambda$ for all $\lambda$.

(b) When $\lambda = 1$, find a basis for $\operatorname{Nul} A_1$.

(c) Determine when the map $\boldsymbol{x} \mapsto A_\lambda \boldsymbol{x}$ is one-to-one, onto and invertible.

(d) Could a $4 \times 7$ matrix have a two-dimensional null space? Explain.

# Solution to Problem 6: Rank and Nullity

Use the parameter-free row replacements

$$R_3 \leftarrow R_3 - R_1, \qquad R_3 \leftarrow R_3 - R_2.$$

Then

$$A_\lambda \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & \lambda - 1 \end{bmatrix}.$$

Therefore

| | $\operatorname{rank} A_\lambda$ | $\dim \operatorname{Nul} A_\lambda$ |
|---|---|---|
| $\lambda \neq 1$ | 3 | 0 |
| $\lambda = 1$ | 2 | 1 |

For $\lambda = 1$,

$$x_1 + x_2 = 0, \qquad x_2 + x_3 = 0.$$

Thus

$$\boxed{\operatorname{Nul} A_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}}.$$

# Solution to Problem 6: Consequences

For $\lambda \neq 1$, the matrix has three pivot positions. By the Invertible Matrix Theorem,

$$\boxed{A_\lambda \text{ is invertible}},$$

and the map $\boldsymbol{x} \mapsto A_\lambda \boldsymbol{x}$ is both one-to-one and onto $\mathbb{R}^3$.
For $\lambda = 1$, the null space contains a nonzero vector, so the map is not one-to-one. Since the matrix is square and has rank $2 < 3$, it is also not onto and is not invertible.
For a $4 \times 7$ matrix $B$, rank–nullity gives

$$\operatorname{rank} B + \dim \operatorname{Nul} B = 7.$$

If $\dim \operatorname{Nul} B = 2$, then $\operatorname{rank} B = 5$, but

$$\operatorname{rank} B \leq 4$$

because $B$ has only four rows. Hence

$$\boxed{\text{a } 4 \times 7 \text{ matrix cannot have nullity } 2}.$$